Abhishek Hegade K R

Contact Information

E-mail: ah4278@princeton.edu, abhikumbale@gmail.com
Citizenship: Indian
Profiles: Personal website, Inspires, Google scholar, github and Linkedin.

Professional Experience

Postdoctoral Fellow-Princeton Gravity Initiative, 2025 -
Princeton University

Education

PhD in Physics, University of Illinois Urbana Champaign 2020 - 2025
Advisor : Nicolás Yunes, GPA : 4.0/4.0.
Teaching/Research Assistant alternating semesters.

BSc (Hons.) in Mathematics and Physics, Chennai Mathematical Institute 2017 - 2020
Advisors : K.G. Arun and Alok Laddha, GPA : 9.52/10.0.
Gold medal awarded for graduating top of my class.

Research Interests

I am currently interested in modeling tidal interactions in neutron star binary systems, understanding EMRI-accretion disk interactions and modeling plasma around supermassive black holes. In the past, I have worked on understanding the breakdown of effective field theories of gravity, the mathematical properties of black holes in and outside general relativity and measuring the Hubble constant using gravitational wave observations.

Awards

Illinois Center for Advanced Studies of the Universe - Physics Fellowship	Fall 2024 - Spring 2025
Scott Anderson Award, University of Illinois Urbana Champaign	Spring 2023
University Fellowship, University of Illinois Urbana Champaign	Fall 2022
Medal of Excellence, Chennai Mathematical Institute	2020
KVPY Fellow, Department of Science and Technology of the Government of India	2017-2020
Indian Academy of Sciences, Summer Research Fellowship	Summer 2019
National Talent Search Examination - State Scholar	2015

Mentoring Experience

- Taeho Kim: Modeling nonlinear tidal effects in neutron star binaries. Princeton Senior thesis (2025-2026).
- Jayana Saes: Universal relationship with dynamical tides (2024-present).
- Anand Balivada: The effect of rotation in post-Newtonian theory of tidal interactions (2024-present).

Talks and workshops

Invited talks and workshops

1. Modeling dynamical tidal interactions in general relativity, Penn State University, April 2026.
2. Relativistic Disk-EMRI Interactions (online), SISSA GW Group Meeting, December 2025.
3. Relativistic Disk-EMRI Interactions, Princeton Gravity Initiative, Princeton, September 2025.

4. Modeling dynamical tidal interactions in general relativity, From Colliders to the Cosmos, Institute for nuclear theory, University of Washington, Seattle, September 2025.
5. Modeling dynamical tidal interactions in general relativity (online), Gravitational Physics Seminars, IIT Gandhinagar, July 2025.
6. Dynamical tides in neutron star binaries, University of Illinois, Urbana-Champaign, May 2025.
7. Out-of-equilibrium tidal effects in gravitational wave observations of binary neutron stars, MIT GRITTS Seminar, MIT, October 2024.
8. Modeling tidal dissipation in neutron star binary systems, Prof. Emanuele Berti's group meeting, Johns Hopkins University, October 2024.
9. Probing dissipative effects in neutron stars using gravitational waves, ICTS Astrophysics Seminar, International Centre for Theoretical Sciences, January 2024.
10. Probing dissipative effects in neutron stars using gravitational waves, VandyGRAF Seminar, Vanderbilt University, November 2023.
11. Advances in Chern-Simons Classical and Quantum Gravity, ICERM, Brown University, May 2022.

Contributed talks

1. Modeling tidal dissipation in neutron star, APS April meeting, Anaheim, April 2025.
2. Probing internal dissipative processes of neutron stars with gravitational waves - II, APS April meeting, Sacramento, April 2024.
3. Where and why does Einstein-Scalar-Gauss-Bonnet theory breakdown? APS April meeting, Minneapolis, April 2023.
4. How Do Black Holes Grow Hair? APS April meeting, New York, April 2022.
5. How Do Black Holes Grow Hair? Midwest Relativity meeting, Urbana-Champaign, November 2021.

Publications

Total 17 papers with 12 as first author and 5 as second author. Student led papers are marked with *.

Under review:

1. **Abhishek Hegade K. R.**, K.J. Kwon, Tejaswi Venumadhav, Hang Yu, and Nicolás Yunes, "The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response".
2. *Anand Balivada, **Abhishek Hegade K. R.**, Nicolás Yunes, "A New Spin on Dissipative Tides: First-Post-Newtonian Effects in Compact Binary Inspirals".
3. **Abhishek Hegade K. R.**and James M. Stone, "Kinetic magnetohydrodynamics and Landau fluid closure in relativity".

Published:

1. **Abhishek Hegade K. R.**, Yumu Yang, Mauricio Hippert, Jacquelyn Noronha-Hostler, Jorge Noronha and Nicolás Yunes, "Dynamical tidal response of neutron stars as a probe of dense-matter properties", accepted in PRD.
2. *Jayana A. Saes, **Abhishek Hegade K. R.**, Nicolás Yunes, "Universal relations with dynamical tides", Phys. Rev. D 113, 104045.
3. **Abhishek Hegade K. R.**, Charles F. Gammie and Nicolás Yunes, "Relativistic treatment of accretion disk torques on extreme mass-ratio inspirals around spinning black holes", Phys.Rev.D 112 (2025) 12, 124068
4. **Abhishek Hegade K. R.**, Charles F. Gammie and Nicolás Yunes, "Relativistic treatment of accretion disk torques on extreme mass-ratio inspirals around nonspinning black holes", Phys. Rev. D 112, 124012
5. **Abhishek Hegade K. R.**, K.J. Kwon, Tejaswi Venumadhav, Hang Yu, and Nicolás Yunes, "Relativistic and Dynamical Love Numbers", Phys.Rev.Lett. 136 (2026)
6. **Abhishek Hegade K. R.**, Justin L. Ripley and Nicolás Yunes, "Dissipative tidal effects to next-to-leading order and constraints on the dissipative tidal deformability using gravitational wave data", Phys. Rev. D 110, 044041, arXiv:2407.02584 [gr-qc].
7. Justin L. Ripley, **Abhishek Hegade K. R.**, Rohit S. Chandramouli and Nicolás Yunes, "First constraint on the dissipative tidal deformability of neutron stars", Nature Astronomy (2024), arXiv:2312.11659 [gr-qc]..
8. **Abhishek Hegade K. R.**, Justin L. Ripley and Nicolás Yunes, "Dynamical tidal response of non-rotating relativistic stars", Phys. Rev. D 109, 104064, arXiv:2403.03254 [gr-qc]
9. Justin L. Ripley, **Abhishek Hegade K. R.**, and Nicolás Yunes, "Probing internal dissipative processes of neutron stars with gravitational waves during the inspiral of neutron star binaries", Phys. Rev. D 108, 103037, arXiv:2306.15633 [gr-qc].
10. **Abhishek Hegade K. R.**, Justin L. Ripley, and Nicolás Yunes, "The non-relativistic limit of first-order relativistic viscous fluids", Phys. Rev. D 107, 124029, arXiv:2305.09725 [gr-qc].
11. **Abhishek Hegade K. R.**, Elias R. Most, Jorge Noronha, Helvi Witek, and Nicolás Yunes, "How Do Axisymmetric Black Holes Grow Monopole and Dipole Hair?", Phys. Rev. D 107, 104047, arXiv:2212.02039 [gr-qc].
12. **Abhishek Hegade K. R.**, Justin L. Ripley, and Nicolás Yunes, "Where and why does Einstein-scalar-Gauss-Bonnet theory break down?" Phys. Rev. D 107, 044044, arXiv:2211.08477 [gr-qc].
13. **Abhishek Hegade K. R.**, Elias R. Most, Jorge Noronha, Helvi Witek, and Nicolás Yunes, "How do spherical black holes grow monopole hair?" Phys. Rev. D 105, 064041, arXiv:2201.055178 [gr-qc].

14. [1]Deep Chatterjee, **Abhishek Hegade K. R.**, Gilbert Holder, Daniel E. Holz, Scott Perkins, Kent Yagi, and Nicolás Yunes, "Cosmology with Love: Measuring the Hubble constant using neutron star universal relations," Phys. Rev. D 104, 083528, arXiv:2106.06589 [gr-qc].

[1]Editor's Suggestion